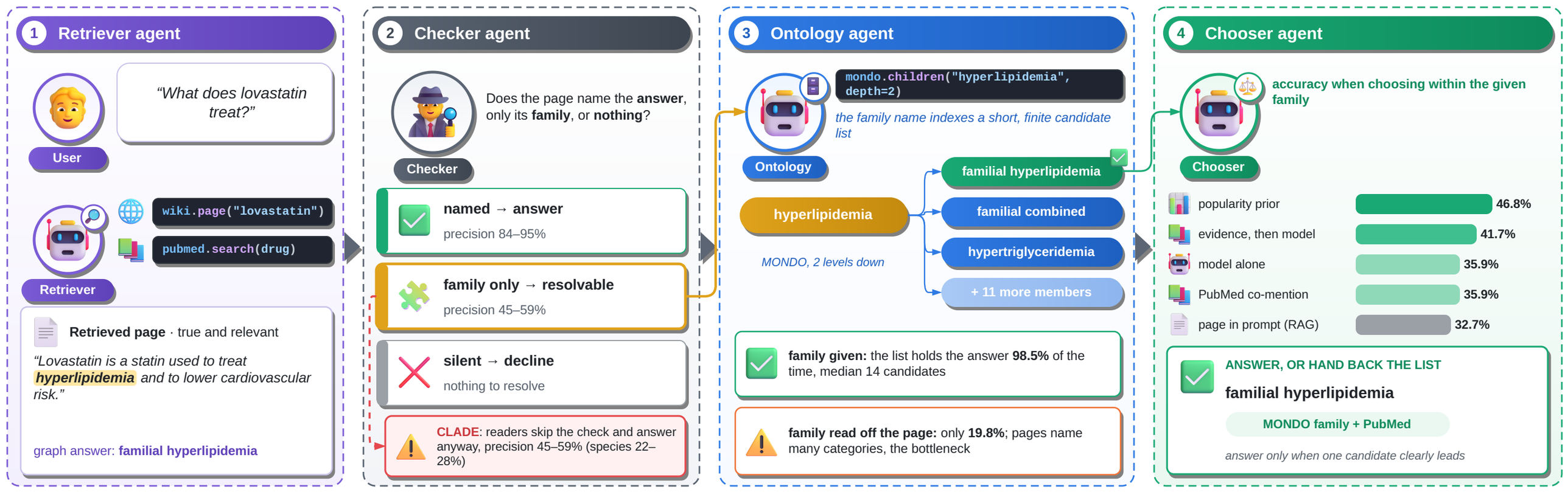

1 Retriever agent
User
“What does lovastatin treat?”
Retriever
wiki.page("lovastatin")
pubmed.search(drug)
Retrieved page · true and relevant
“Lovastatin is a statin used to treat hyperlipidemia and to lower cardiovascular risk.”
graph answer: familial hyperlipidemia
2 Checker agent
Checker
Does the page name the answer, only its family, or nothing?
named → answer
precision 84–95%
family only → resolvable
precision 45–59%
silent → decline
nothing to resolve
CLADE: readers skip the check and answer anyway, precision 45–59% (species 22–28%)
3 Ontology agent
Ontology
mondo.children("hyperlipidemia", depth=2)
the family name indexes a short, finite candidate list
hyperlipidemia
MONDO, 2 levels down
familial hyperlipidemia
familial combined
hypertriglyceridemia
+ 11 more members
family given: the list holds the answer 98.5% of the time, median 14 candidates
family read off the page: only 19.8%; pages name many categories, the bottleneck
4 Chooser agent
Chooser
accuracy when choosing within the given family
popularity prior 46.8%
evidence, then model 41.7%
model alone 35.9%
PubMed co-mention 35.9%
page in prompt (RAG) 32.7%
ANSWER, OR HAND BACK THE LIST
familial hyperlipidemia
MONDO family + PubMed
answer only when one candidate clearly leads